

Esercizio Malware

Pratica S9/L2 | Analisi Statica e Dinamica | notepad-classico.exe | FlareVM + PE-bear + ProcMon

Sezione 1

Obiettivo dell'esercizio

L'esercizio ha richiesto l'analisi del file eseguibile notepad-classico.exe presente nella cartella Malware della macchina virtuale FlareVM. L'obiettivo era identificare le librerie importate e le sezioni PE del campione, fornendo una descrizione per ciascuna tramite analisi assistita da AI, con integrazione di analisi dinamica per osservare il comportamento a runtime.

Gli strumenti utilizzati sono stati PE-bear v0.7.1 per l'analisi statica (imports, sections, strings, disassembly) e Process Monitor di Sysinternals per la cattura degli eventi a runtime.

Sezione 2

Campione e ambiente di analisi

Parametro	Valore
File analizzato	notepad-classico.exe
Dimensione	283 KB
Percorso	C:\Users\FlareVm\Desktop\Malware\notepad-classico.exe
Architettura	x86 (32-bit) – eseguito via WOW64 su sistema x64
Ambiente di analisi	FlareVM su Windows 10 x64 – VirtualBox Host-Only
Strumento statico	PE-bear v0.7.1
Strumento dinamico	Process Monitor (Sysinternals)
Snapshot baseline	Clean – rollback eseguito post-analisi

Sezione 3

Analisi statica — PE-bear

Ho caricato il campione in PE-bear e analizzato la struttura PE completa. L'analisi ha rivelato anomalie strutturali significative che indicano la presenza di codice malevolo nascosto dietro una facciata di Notepad legittimo.

3.1 Librerie importate

Il tab Imports mostra 9 DLL nella Import Address Table statica per un totale di 201 funzioni. La presenza di LoadLibraryA e GetProcAddress in KERNEL32.dll indica che il campione carica librerie aggiuntive dinamicamente a runtime, non dichiarate nella IAT statica.

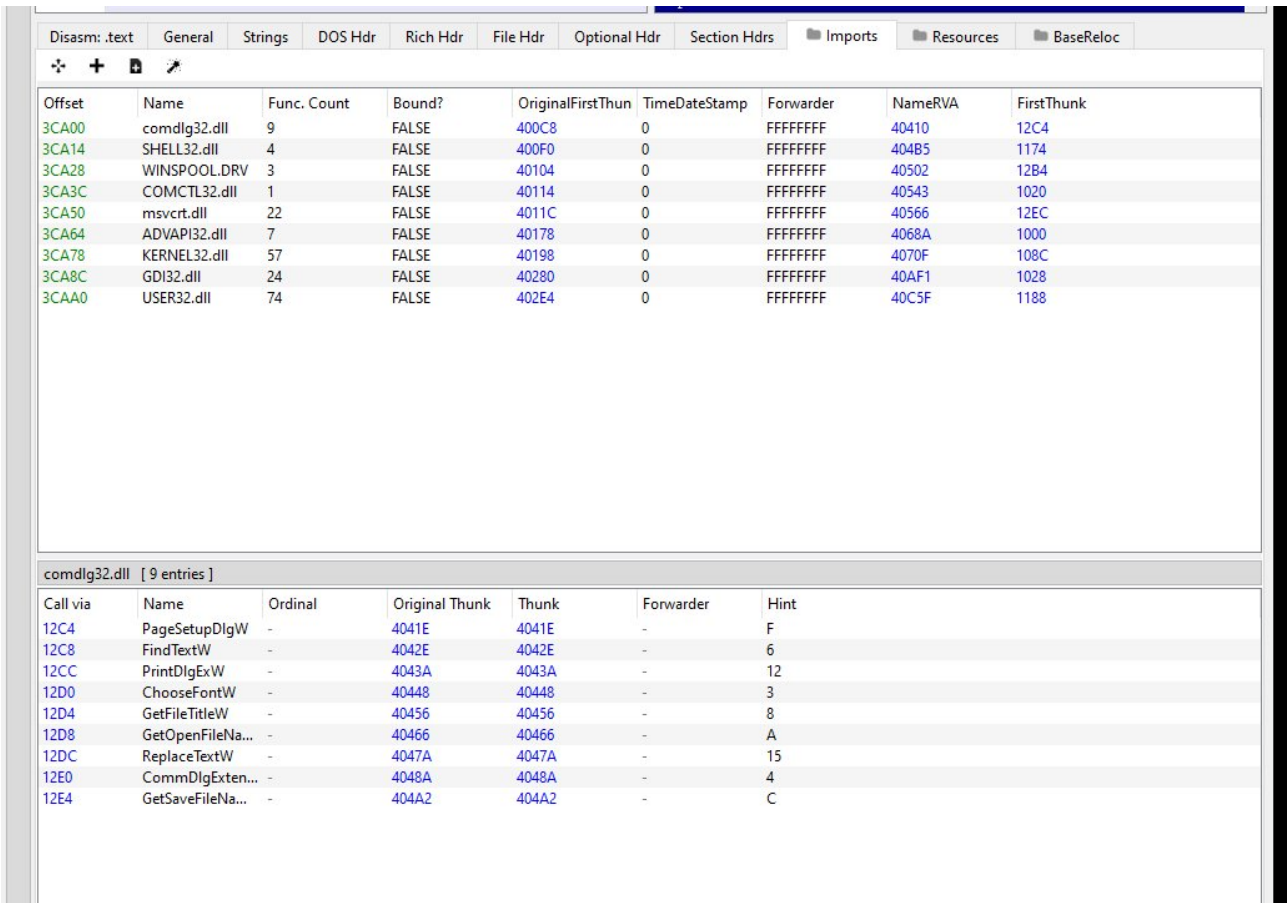


Fig. 1 — Tab Imports PE-bear: 9 DLL con conteggio funzioni e dettaglio comdlg32.dll.

Libreria	Func.	Funzioni chiave	Valutazione
USER32.dll	74	Gestione finestre, messaggi, input, hook	Legittima per GUI — 74 funzioni è elevato
KERNEL32.dll	57	LoadLibraryA, GetProcAddress, CreateThread, ReadFile	Sospetta — import dinamici a runtime
GDI32.dll	24	Rendering grafico, font, BitBlt	Legittima per editor testo
msvcrt.dll	22	Runtime C: malloc, memcpy, printf	Standard — runtime C
comdlg32.dll	9	GetOpenFileNameW, PrintDlgExW, FindTextW, ReplaceTextW	Legittima — dialoghi file standard
ADVAPI32.dll	7	RegOpenKey, RegQueryValue, CryptGenRandom	Da monitorare — accesso registry e crypto
SHELL32.dll	4	ShellExecuteW, DragAcceptFiles, DragQueryFileW	Potenzialmente sospetta — esecuzione processi
WINSPOOL.DRV	3	OpenPrinter, GetPrinterData, DocumentProperties	Legittima — stampa documenti
COMCTL32.dll	1	InitCommonControlsEx	Legittima — controlli UI avanzati

Il red flag principale è `KERNEL32.dll` con `LoadLibraryA`, `GetProcAddress` e `CreateThread`: pattern classico di reflective loading. Il malware ricostruisce la IAT dinamicamente per nascondere le dipendenze reali all'analisi statica.

3.2 Sezioni PE

Il tab Section Hdrs rivela 6 sezioni con due anomalie critiche: due sezioni con nome .text e due con nome .rsrc — struttura atipica per un eseguibile legittimo.

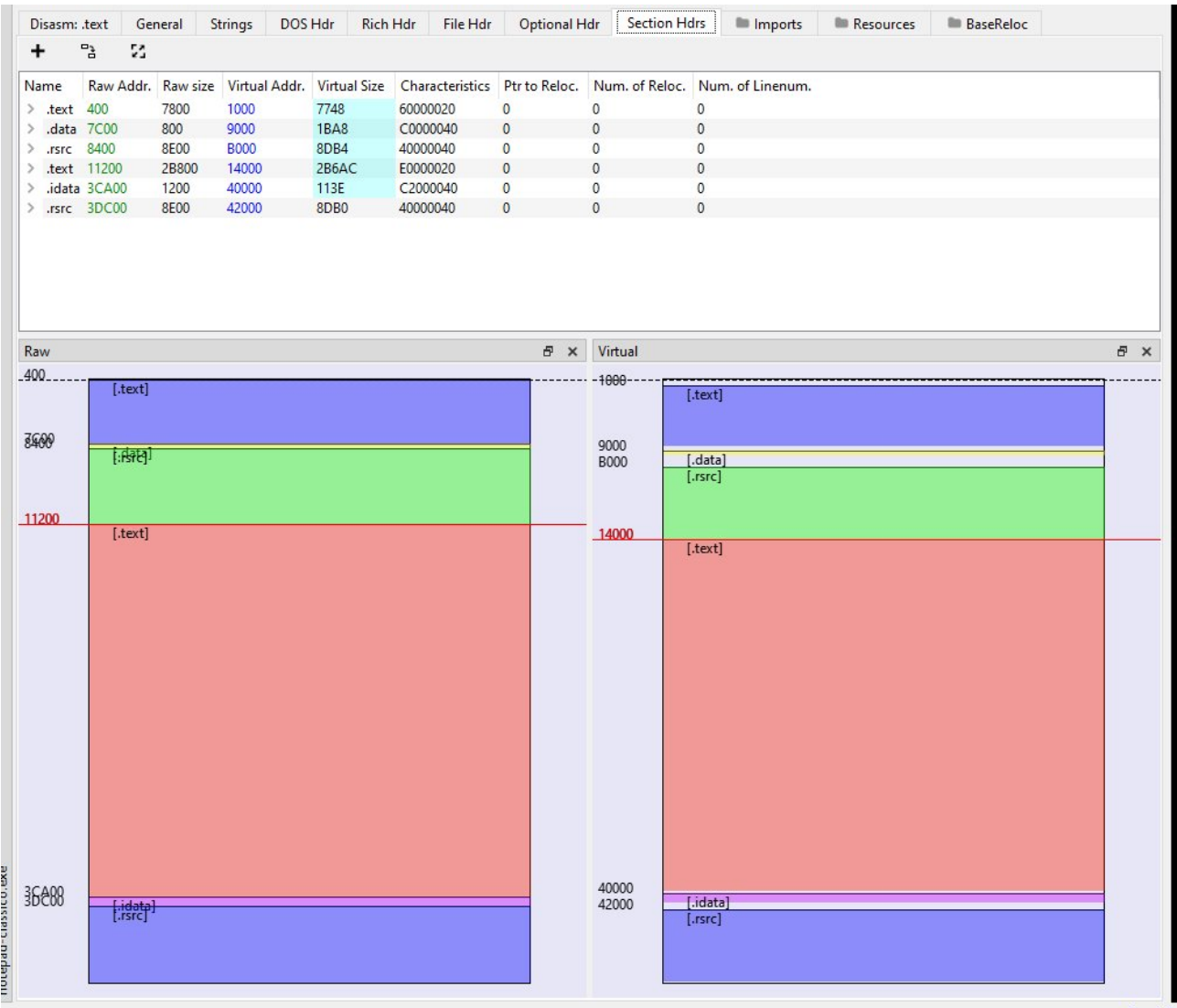


Fig. 2 — Tab Section Hdrs PE-bear: 6 sezioni con layout Raw e Virtual. La seconda .text (rosa) domina lo spazio.

Nome	Raw Addr	Virt Addr	Virt Size	Caratteristiche	Analisi
.text	0x400	0x1000	0x7748	60000020	Codice Notepad legittimo — Execute+Read
.data	0x7C00	0x9000	0x1BA8	C0000040	Dati inizializzati — Read+Write
.rsrc	0x8400	0xB000	0x8DB4	40000040	Risorse UI — icone, stringhe
.text	0x11200	0x14000	0x2B6AC	E0000020	ANOMALA — Exec+Read+Write — payload
.idata	0x3CA00	0x40000	0x113E	C2000040	Import table
.rsrc	0x3DC00	0x42000	0x8DB0	40000040	DUPLICATA — possibile payload cifrato

La seconda sezione .text (offset 0x11200) è l'anomalia principale: caratteristiche E0000020 — Execute+Read+Write simultaneamente. Una sezione scrivibile ed eseguibile è il pattern tipico di payload

unpacked a runtime. L'Entry Point punta a 0x11200, confermando che l'esecuzione parte dalla sezione anomala.

3.3 Analisi stringhe

PE-bear ha estratto 2235 stringhe. L'analisi ha identificato tre zone distinte: stringhe Notepad legittime (confermate da notepad.pdb), sequenze di opcode x86 lette come ASCII dalla seconda .text, e nomi DLL hardcoded per la ricostruzione dinamica della IAT.

Extracted strings: 2235

Save Page 0 Max per page 10000 Search string ☐ By regex ☐ Case sensitive

	Offset	Type	Length	String
35	a44	W	8	IfItalic
36	a58	W	8	IfWeight
37	a6c	W	13	IfOrientation
38	a88	W	12	IfEscapement
39	aa4	W	26	Software\Microsoft\Notepad
40	adc	W	14	Lucida Console
41	afc	W	24	Out of RC string space!!
42	b30	W	10	DEV Error!
43	b48	W	9	SlipUpAcc
44	b5c	W	7	./SETUP
45	ba0	W	7	MainAcc
46	bb0	W	12	commdlg_help
47	bcc	W	19	commdlg_FindReplace
48	c48	A	10	hhctrl.ocx
49	c54	A	59	CLSID\{ADB880A6-D8FF-11CF-9377-00AA003B7A11}\InprocServer32
50	cf0	A	5	RSDS,
51	d08	A	11	notepad.pdb
52	fb0	A	5	t PPh
53	15ef	A	7	HtIHt -
54	1fff	A	7	Ot Ot23
55	2262	A	5	t;Ot(
56	2736	A	6	u/VVWQ
57	28ce	A	5	HtIHu
58	3373	A	5	GGCPC
59	33ed	A	6	AABFF3
60	3555	A	5	@_[^]
61	365b	A	5	j1Vt)
62	3773	A	5	uF98
63	3b63	A	5	0PPVh
64	3e20	A	5	uFj3W
65	3f87	A	6	AA@@Nu
66	4054	A	5	tN9=X

Fig. 3 — Strings tab: zona shellcode — opcode x86 ASCII (PVVVS, SVSWH) + nomi DLL hardcoded (comdlg32.dll, SHELL32.dll).

Zona	Offset	Tipo	Significato
Notepad legittimo	0x7a0 – 0xd08	Unicode W	NpEncodingDialog, notepad.pdb, iWindowPosX — Notepad autentico ricompilato
Shellcode	0xfbc – 0x6902	ASCII opcode x86	PVVVS, u/VVWQ, SVSWH, j XPVSS — sequenze PUSH/POP/CALL della seconda .text
Import dinamici	0x6e16 – 0x6efa	ASCII DLL names	comdlg32.dll, SHELL32.dll, GetSaveFileNameW — IAT ricostruita a runtime

Sezione 4

Analisi dinamica — Process Monitor

Ho eseguito notepad-classico.exe nella FlareVM con Process Monitor attivo e filtro su Process Name is notepad-classico.exe, catturando il comportamento completo dall'avvio fino alle operazioni di file system.

4.1 Comportamento visivo

All'esecuzione appare un Blocco note vuoto (Senza nome - Blocco note) perfettamente funzionante e indistinguibile da un Notepad legittimo. L'utente non percepisce alcuna anomalia visiva mentre il payload opera silenziosamente in background.

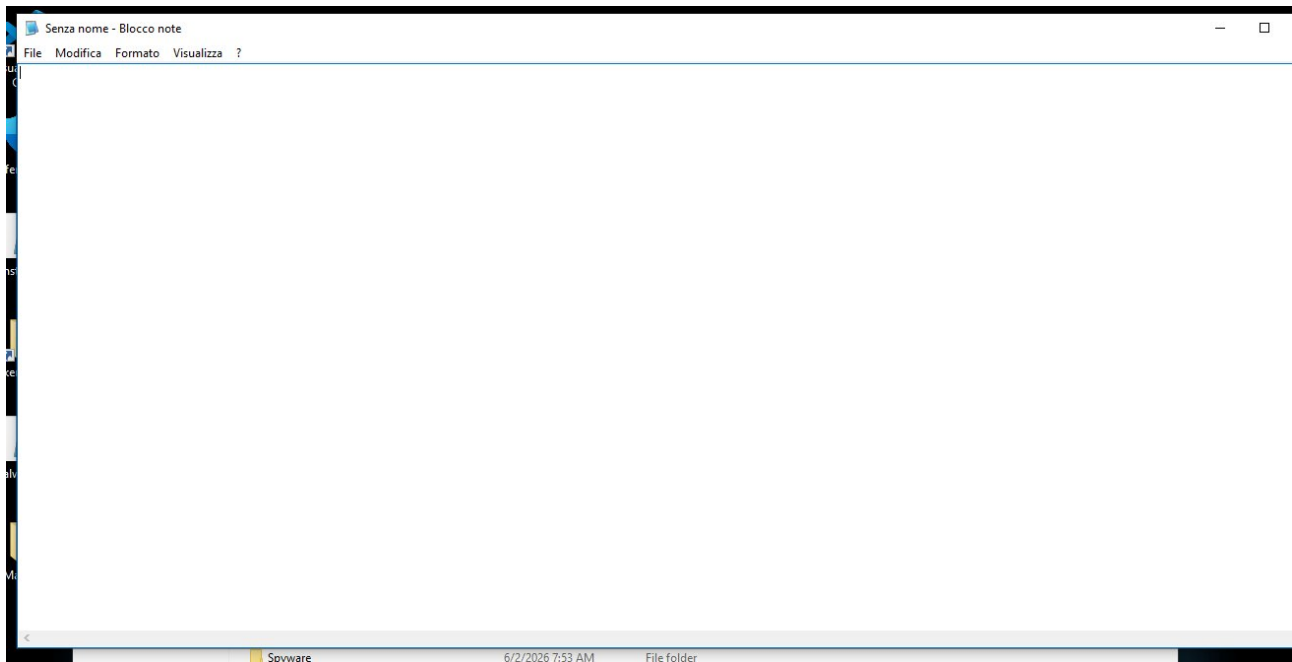


Fig. 4 — Comportamento visivo: Blocco note vuoto all'esecuzione — nessuna anomalia visibile all'utente.

4.2 ProcMon — fase 1: avvio e registry scan

Nella prima fase il campione attiva il layer WOW64, carica kernel32 dinamicamente e avvia una scansione massiva del registro di sistema con pattern anti-VM.

Process Monitor - Sysinternals: www.sysinternals.com

File Edit Event Filter Tools Options Help

Time ...	Process Name	PID	Operation	Path	Result	Detail
9:00:5...	notepad-classic...	4364	Process Start		SUCCESS	Parent PID: 3324 ...
9:00:5...	notepad-classic...	4364	Thread Create		SUCCESS	Thread ID: 4528
9:00:5...	notepad-classic...	4364	Load Image	C:\Users\FlareVm\Desktop\Malware\n...	SUCCESS	Image Base: 0x100...
9:00:5...	notepad-classic...	4364	Load Image	C:\Windows\System32\ntdll.dll	SUCCESS	Image Base: 0x7ff9...
9:00:5...	notepad-classic...	4364	Load Image	C:\Windows\SysWOW64\ntdll.dll	SUCCESS	Image Base: 0x775...
9:00:5...	notepad-classic...	4364	RegOpenKey	HKLM\System\CurrentControlSet\Contr...	REPARSE	Desired Access: Q...
9:00:5...	notepad-classic...	4364	RegOpenKey	HKLM\System\CurrentControlSet\Contr...	SUCCESS	Desired Access: Q...
9:00:5...	notepad-classic...	4364	RegQueryValue	HKLM\System\CurrentControlSet\Contr...	NAME NOT FOUND	Length: 80
9:00:5...	notepad-classic...	4364	RegCloseKey	HKLM\System\CurrentControlSet\Contr...	SUCCESS	
9:00:5...	notepad-classic...	4364	RegOpenKey	HKLM\SYSTEM\CurrentControlSet\Con...	REPARSE	Desired Access: Q...
9:00:5...	notepad-classic...	4364	RegOpenKey	HKLM\System\CurrentControlSet\Contr...	NAME NOT FOUND	Desired Access: Q...
9:00:5...	notepad-classic...	4364	RegOpenKey	HKLM\SYSTEM\CurrentControlSet\Con...	REPARSE	Desired Access: Q...
9:00:5...	notepad-classic...	4364	RegOpenKey	HKLM\System\CurrentControlSet\Contr...	SUCCESS	Desired Access: Q...
9:00:5...	notepad-classic...	4364	RegQueryValue	HKLM\System\CurrentControlSet\Contr...	NAME NOT FOUND	Length: 24
9:00:5...	notepad-classic...	4364	RegCloseKey	HKLM\System\CurrentControlSet\Contr...	SUCCESS	
9:00:5...	notepad-classic...	4364	CreateFile	C:\Windows	SUCCESS	Desired Access: E...
9:00:5...	notepad-classic...	4364	Load Image	C:\Windows\System32\wow64.dll	SUCCESS	Image Base: 0x7ff9...
9:00:5...	notepad-classic...	4364	Load Image	C:\Windows\System32\wow64win.dll	SUCCESS	Image Base: 0x7ff9...
9:00:5...	notepad-classic...	4364	CreateFile	C:\Windows\System32\wow64log.dll	NAME NOT FOUND	Desired Access: R...
9:00:5...	notepad-classic...	4364	CreateFile	C:\Windows	SUCCESS	Desired Access: R...
9:00:5...	notepad-classic...	4364	QueryNameInfo...	C:\Windows	SUCCESS	Name: \Windows
9:00:5...	notepad-classic...	4364	CloseFile	C:\Windows	SUCCESS	
9:00:5...	notepad-classic...	4364	RegOpenKey	HKLM\Software\Microsoft\Wow64\x86	SUCCESS	Desired Access: R...
9:00:5...	notepad-classic...	4364	RegQueryValue	HKLM\SOFTWARE\Microsoft\Wow64\...	NAME NOT FOUND	Length: 520
9:00:5...	notepad-classic...	4364	RegQueryValue	HKLM\SOFTWARE\Microsoft\Wow64\...	SUCCESS	Type: REG_SZ, Le...
9:00:5...	notepad-classic...	4364	RegCloseKey	HKLM\SOFTWARE\Microsoft\Wow64\...	SUCCESS	
9:00:5...	notepad-classic...	4364	Load Image	C:\Windows\System32\wow64cpu.dll	SUCCESS	Image Base: 0x775...
9:00:5...	notepad-classic...	4364	RegOpenKey	HKLM\System\CurrentControlSet\Contr...	REPARSE	Desired Access: Q...
9:00:5...	notepad-classic...	4364	RegOpenKey	HKLM\System\CurrentControlSet\Contr...	SUCCESS	Desired Access: Q...
9:00:5...	notepad-classic...	4364	RegSetInfoKey	HKLM\System\CurrentControlSet\Contr...	SUCCESS	KeySetInformation...
9:00:5...	notepad-classic...	4364	RegQueryValue	HKLM\System\CurrentControlSet\Contr...	NAME NOT FOUND	Length: 80
9:00:5...	notepad-classic...	4364	RegCloseKey	HKLM\System\CurrentControlSet\Contr...	SUCCESS	
9:00:5...	notepad-classic...	4364	RegOpenKey	HKLM\SYSTEM\CurrentControlSet\Con...	REPARSE	Desired Access: Q...
9:00:5...	notepad-classic...	4364	RegOpenKey	HKLM\System\CurrentControlSet\Contr...	NAME NOT FOUND	Desired Access: Q...
9:00:5...	notepad-classic...	4364	RegOpenKey	HKLM\SYSTEM\CurrentControlSet\Con...	REPARSE	Desired Access: Q...
9:00:5...	notepad-classic...	4364	RegOpenKey	HKLM\System\CurrentControlSet\Contr...	SUCCESS	Desired Access: Q...
9:00:5...	notepad-classic...	4364	RegSetInfoKey	HKLM\System\CurrentControlSet\Contr...	SUCCESS	KeySetInformation...
9:00:5...	notepad-classic...	4364	RegQueryValue	HKLM\System\CurrentControlSet\Contr...	NAME NOT FOUND	Length: 24
9:00:5...	notepad-classic...	4364	RegCloseKey	HKLM\System\CurrentControlSet\Contr...	SUCCESS	
9:00:5...	notepad-classic...	4364	CreateFile	C:\Users\FlareVm\Desktop\Malware	SUCCESS	Desired Access: E...
9:00:5...	notepad-classic...	4364	Load Image	C:\Windows\SysWOW64\kernel32.dll	SUCCESS	Image Base: 0x770...
9:00:5...	notepad-classic...	4364	Load Image	C:\Windows\SysWOW64\KernelBase.dll	SUCCESS	Image Base: 0x76c...
9:00:5...	notepad-classic...	4364	RegQueryValue	HKLM\System\CurrentControlSet\Contr...	NAME NOT FOUND	Length: 528
9:00:5...	notepad-classic...	4364	QueryNameInfo...	C:\Windows\SysWOW64\KernelBase.dll	SUCCESS	Name: \Windows\...
9:00:5...	notepad-classic...	4364	RegOpenKey	HKLM\System\CurrentControlSet\Contr...	REPARSE	Desired Access: R...
9:00:5...	notepad-classic...	4364	RegOpenKey	HKLM\System\CurrentControlSet\Contr...	NAME NOT FOUND	Desired Access: R...
9:00:5...	notepad-classic...	4364	RegQueryValue	HKLM\System\CurrentControlSet\Contr...	NAME NOT FOUND	Length: 528
9:00:5...	notepad-classic...	4364	QueryNameInfo...	C:\Windows\SysWOW64\KernelBase.dll	SUCCESS	Name: \Windows\...
9:00:5...	notepad-classic...	4364	RegOpenKey	HKLM\System\CurrentControlSet\Contr...	REPARSE	Desired Access: Q...
9:00:5...	notepad-classic...	4364	RegOpenKey	HKLM\System\CurrentControlSet\Contr...	NAME NOT FOUND	Desired Access: Q...
9:00:5...	notepad-classic...	4364	RegOpenKey	HKLM\System\CurrentControlSet\Contr...	REPARSE	Desired Access: R...
9:00:5...	notepad-classic...	4364	RegOpenKey	HKLM\System\CurrentControlSet\Contr...	NAME NOT FOUND	Desired Access: R...
9:00:5...	notepad-classic...	4364	RegOpenKey	HKLM\Software\WOW6432Node\Polic...	REPARSE	Desired Access: Q...

Fig. 5 — ProcMon fase 1: Process Start, Thread Create, caricamento WOW64 e scansione HKLM\System.

Evento	Dettaglio	Analisi
Process Start	PID 4364 — Parent PID 3324 (Explorer)	Avvio regolare da Explorer
Thread Create	Thread ID 4528 — immediato all'avvio	Thread secondario creato subito — launcher payload
Load Image	wow64.dll, wow64win.dll, wow64cpu.dll	Layer WOW64 attivato — esecuzione 32-bit su x64
Load Image	SysWOW64\kernel32.dll, KernelBase.dll	Caricamento dinamico kernel32 — confermato da analisi statica

Evento	Dettaglio	Analisi
RegQueryValue	HKLM\System\CurrentControlSet — NAME NOT FOUND	Cerca chiavi inesistenti — possibile check anti-VM/sandbox
RegSetInfoKey	HKLM\System\CurrentControlSet — SUCCESS	Scrittura REG_DWORD nel registro di sistema confermata

4.3 ProcMon — fase 2: DLL dinamiche e filesystem

Nella seconda fase il campione completa la ricostruzione della IAT caricando le DLL trovate come stringhe hardcoded nell'analisi statica, crea un secondo thread e avvia operazioni anomale sul filesystem.

Process Monitor - Sysinternals: www.sysinternals.com

Time ...	Process Name	PID	Operation	Path	Result	Detail
9:00:5...	notepad-classic...	4364	RegQueryValue	HKLM\System\CurrentControlSet\Contr...	NAME NOT FOUND	Length: 528
9:00:5...	notepad-classic...	4364	QueryNameInfo...	C:\Windows\SysWOW64\KernelBase.dll	SUCCESS	Name: \Windows\...
9:00:5...	notepad-classic...	4364	RegOpenKey	HKLM\System\CurrentControlSet\Contr...	REPARSE	Desired Access: R...
9:00:5...	notepad-classic...	4364	RegOpenKey	HKLM\System\CurrentControlSet\Contr...	NAME NOT FOUND	Desired Access: R...
9:00:5...	notepad-classic...	4364	RegQueryValue	HKLM\System\CurrentControlSet\Contr...	NAME NOT FOUND	Length: 528
9:00:5...	notepad-classic...	4364	QueryNameInfo...	C:\Windows\SysWOW64\KernelBase.dll	SUCCESS	Name: \Windows\...
9:00:5...	notepad-classic...	4364	RegOpenKey	HKLM\System\CurrentControlSet\Contr...	REPARSE	Desired Access: Q...
9:00:5...	notepad-classic...	4364	RegOpenKey	HKLM\System\CurrentControlSet\Contr...	NAME NOT FOUND	Desired Access: Q...
9:00:5...	notepad-classic...	4364	RegOpenKey	HKLM\System\CurrentControlSet\Contr...	REPARSE	Desired Access: R...
9:00:5...	notepad-classic...	4364	RegOpenKey	HKLM\System\CurrentControlSet\Contr...	NAME NOT FOUND	Desired Access: R...
9:00:5...	notepad-classic...	4364	RegOpenKey	HKLM\Software\WOW6432Node\Polic...	REPARSE	Desired Access: Q...
9:00:5...	notepad-classic...	4364	RegOpenKey	HKLM\SOFTWARE\Policies\Microsoft\...	SUCCESS	Desired Access: Q...
9:00:5...	notepad-classic...	4364	RegSetInfoKey	HKLM\SOFTWARE\Policies\Microsoft\...	SUCCESS	KeySetInformation...
9:00:5...	notepad-classic...	4364	RegQueryValue	HKLM\SOFTWARE\Policies\Microsoft\...	NAME NOT FOUND	Length: 80
9:00:5...	notepad-classic...	4364	RegCloseKey	HKLM\SOFTWARE\Policies\Microsoft\...	SUCCESS	
9:00:5...	notepad-classic...	4364	RegOpenKey	HKCU\Software\Policies\Microsoft\Win...	NAME NOT FOUND	Desired Access: Q...
9:00:5...	notepad-classic...	4364	RegOpenKey	HKLM\System\CurrentControlSet\Contr...	REPARSE	Desired Access: R...
9:00:5...	notepad-classic...	4364	RegOpenKey	HKLM\System\CurrentControlSet\Contr...	SUCCESS	Desired Access: R...
9:00:5...	notepad-classic...	4364	RegSetInfoKey	HKLM\System\CurrentControlSet\Contr...	SUCCESS	KeySetInformation...
9:00:5...	notepad-classic...	4364	RegQueryValue	HKLM\System\CurrentControlSet\Contr...	SUCCESS	Type: REG_DWO...
9:00:5...	notepad-classic...	4364	RegCloseKey	HKLM\System\CurrentControlSet\Contr...	SUCCESS	
9:00:5...	notepad-classic...	4364	Load Image	C:\Windows\SysWOW64\comdlg32.dll	SUCCESS	Image Base: 0x773...
9:00:5...	notepad-classic...	4364	Load Image	C:\Windows\SysWOW64\msvcrt.dll	SUCCESS	Image Base: 0x76e...
9:00:5...	notepad-classic...	4364	Thread Create		SUCCESS	Thread ID: 3808
9:00:5...	notepad-classic...	4364	Load Image	C:\Windows\SysWOW64\combase.dll	SUCCESS	Image Base: 0x76a...
9:00:5...	notepad-classic...	4364	Load Image	C:\Windows\SysWOW64\urtbase.dll	SUCCESS	Image Base: 0x75b...
9:00:5...	notepad-classic...	4364	Load Image	C:\Windows\SysWOW64\vpport4.dll	SUCCESS	Image Base: 0x771...
9:00:5...	notepad-classic...	4364	Load Image	C:\Windows\SysWOW64\SHCore.dll	SUCCESS	Image Base: 0x768...
9:00:5...	notepad-classic...	4364	Load Image	C:\Windows\SysWOW64\user32.dll	SUCCESS	Image Base: 0x767...
9:00:5...	notepad-classic...	4364	Load Image	C:\Windows\SysWOW64\win32u.dll	SUCCESS	Image Base: 0x770...
9:00:5...	notepad-classic...	4364	RegOpenKey	HKLM\SYSTEM\CurrentControlSet\Con...	REPARSE	Desired Access: Q...
9:00:5...	notepad-classic...	4364	RegOpenKey	HKLM\System\CurrentControlSet\Contr...	SUCCESS	Desired Access: Q...
9:00:5...	notepad-classic...	4364	RegSetInfoKey	HKLM\System\CurrentControlSet\Contr...	SUCCESS	KeySetInformation...
9:00:5...	notepad-classic...	4364	RegQueryValue	HKLM\System\CurrentControlSet\Contr...	NAME NOT FOUND	Length: 24
9:00:5...	notepad-classic...	4364	RegCloseKey	HKLM\System\CurrentControlSet\Contr...	SUCCESS	
9:00:5...	notepad-classic...	4364	Load Image	C:\Windows\SysWOW64\gdi32.dll	SUCCESS	Image Base: 0x771...
9:00:5...	notepad-classic...	4364	Load Image	C:\Windows\SysWOW64\gdi32full.dll	SUCCESS	Image Base: 0x75e...
9:00:5...	notepad-classic...	4364	Load Image	C:\Windows\SysWOW64\msvcp_win.dll	SUCCESS	Image Base: 0x76f...
9:00:5...	notepad-classic...	4364	Load Image	C:\Windows\SysWOW64\shlwapi.dll	SUCCESS	Image Base: 0x754...
9:00:5...	notepad-classic...	4364	RegOpenKey	HKLM\Software\WOW6432Node\Micr...	NAME NOT FOUND	Desired Access: E...
9:00:5...	notepad-classic...	4364	CreateFile	C:\Windows\WinSxS\x86_microsoft.win...	SUCCESS	Desired Access: E...
9:00:5...	notepad-classic...	4364	Load Image	C:\Windows\SysWOW64\shell32.dll	SUCCESS	Image Base: 0x755...
9:00:5...	notepad-classic...	4364	Load Image	C:\Windows\SysWOW64\advapi32.dll	SUCCESS	Image Base: 0x760...
9:00:5...	notepad-classic...	4364	Load Image	C:\Windows\SysWOW64\sechost.dll	SUCCESS	Image Base: 0x75d...
9:00:5...	notepad-classic...	4364	ReadFile	C:\Directory	SUCCESS	Offset: 9,707,520, ...
9:00:5...	notepad-classic...	4364	ReadFile	C:\Directory	SUCCESS	Offset: 9,494,528, ...
9:00:5...	notepad-classic...	4364	ReadFile	C:\Directory	SUCCESS	Offset: 10,047,488, ...
9:00:5...	notepad-classic...	4364	CreateFile	C:\Windows\WinSxS\x86_microsoft.win...	SUCCESS	Desired Access: R...
9:00:5...	notepad-classic...	4364	QueryBasicInfor...	C:\Windows\WinSxS\x86_microsoft.win...	SUCCESS	CreationTime: 12/3...
9:00:5...	notepad-classic...	4364	CloseFile	C:\Windows\WinSxS\x86_microsoft.win...	SUCCESS	
9:00:5...	notepad-classic...	4364	CreateFile	C:\Windows\WinSxS\x86_microsoft.win...	SUCCESS	Desired Access: R...
9:00:5...	notepad-classic...	4364	CreateFileMapp...	C:\Windows\WinSxS\x86_microsoft.win...	FILE LOCKED WI...	SyncType: SyncTy...
9:00:5...	notepad-classic...	4364	CreateFileMapp...	C:\Windows\WinSxS\x86_microsoft.win...	SUCCESS	SyncType: SyncTy...

Fig. 6 — ProcMon fase 2: caricamento comdlg32/shell32/advapi32, Thread Create 3808, ReadFile C:\\$Directory.

Evento	Dettaglio	Analisi
Load Image	SysWOW64\comdlg32.dll, shell32.dll	Confermati da stringhe hardcoded trovate staticamente
Load Image	advapi32.dll, sechost.dll, shlwapi.dll	Accesso registry avanzato e gestione servizi
Thread Create	Thread ID 3808 — dopo caricamento DLL	Payload lanciato in thread separato dopo ricostruzione IAT
RegOpenKey	HKLM\SOFTWARE\Policies\Microsoft — NAME NOT FOUND	Cerca policy di gruppo — possibile bypass UAC o anti-AV
RegSetInfoKey	Type: REG_DWORD — SUCCESS	Seconda scrittura registro confermata
ReadFile	C:\\$Directory — offset 9.4M, 9.7M, 10.0M	Enumerazione MFT NTFS — scansione filesystem per file target
CreateFileMapping	x86_microsoft.win — FILE LOCKED	Tentativo mapping file di sistema — possibile injection

Sezione 5

Considerazione finale

L'analisi statica e dinamica di notepad-classico.exe ha rivelato un trojan dropper con tecnica di camuffamento avanzata. Il campione implementa una strategia in due livelli: una facciata di Notepad completamente funzionante — costruita a partire dal sorgente originale di Windows, come confermato dalla presenza di notepad.pdb — e un payload malevolo nascosto nella seconda sezione .text con caratteristiche Execute+Read+Write.

Il flusso di esecuzione ricostruito è il seguente: l'Entry Point a 0x11200 avvia il loader nella sezione anomala, che attiva WOW64, ricostruisce la IAT a runtime tramite LoadLibraryA/GetProcAddress, scansiona il registro cercando chiavi anti-VM, scrive valori REG_DWORD nel ControlSet, lancia un thread secondario con il payload reale ed enumera il filesystem tramite lettura diretta della MFT NTFS.

Il confronto con Annabelle Ransomware analizzato nella pratica S9/L1 è paradigmatico: Annabelle è aggressivo e visibile — suono, schermata di riscatto, MBR overwrite, impatto immediato e intenzionalmente intimidatorio. notepad-classico.exe adotta l'approccio opposto: massima furtività, nessuna anomalia visiva, azione silente in background. Questo profilo corrisponde a un RAT o spyware il cui obiettivo è la persistenza prolungata e la raccolta di informazioni senza che l'utente si accorga della compromissione.

L'utilizzo dell'AI nell'analisi ha permesso di correlare i tag comportamentali statici con gli eventi ProcMon dinamici, producendo un quadro interpretativo coerente che avrebbe richiesto tempi molto più lunghi con la sola analisi manuale.